High School: Algebra » Seeing Structure in Expressions

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Standards in this domain:

CCSS.MATH.CONTENT.HSA.SSE.A.1	CCSS.MATH.CONTENT.HSA.SSE.A.2	CCSS.MATH.CONTENT.HSA.SSE.B.3
CCSS.MATH.CONTENT.HSA.SSE.B.4		

Interpret the structure of expressions.

CCSS.MATH.CONTENT.HSA.SSE.A.1 (../../../../MATH/CONTENT/HSA/SSE/A/1/)

Interpret expressions that represent a quantity in terms of its context. *

CCSS.MATH.CONTENT.HSA.SSE.A.1.A (../../../../MATH/CONTENT/HSA/SSE/A/1/A/)

Interpret parts of an expression, such as terms, factors, and coefficients.

CCSS.MATH.CONTENT.HSA.SSE.A.1.B (../../../../MATH/CONTENT/HSA/SSE/A/1/B/)

Interpret complicated expressions by viewing one or more of their parts as a single entity. *For example, interpret $P(1+r)^n$ as the product of P and a factor not depending on P.*

CCSS.MATH.CONTENT.HSA.SSE.A.2 (../../../../MATH/CONTENT/HSA/SSE/A/2/)

Use the structure of an expression to identify ways to rewrite it. *For example, see $x^4 - y^4$ as $(x^2)^2 - (y^2)^2$, thus recognizing it as a difference of squares that can be factored as $(x^2 - y^2)(x^2 + y^2)$.*

Write expressions in equivalent forms to solve problems.

CCSS.MATH.CONTENT.HSA.SSE.B.3 (../../../../MATH/CONTENT/HSA/SSE/B/3/)

Choose and produce an equivalent form of an expression to reveal and explain properties of the quantity represented by the expression. *

CCSS.MATH.CONTENT.HSA.SSE.B.3.A (../../../../MATH/CONTENT/HSA/SSE/B/3/A/)

Factor a quadratic expression to reveal the zeros of the function it defines.

CCSS.MATH.CONTENT.HSA.SSE.B.3.B (../../../../MATH/CONTENT/HSA/SSE/B/3/B/)

Complete the square in a quadratic expression to reveal the maximum or minimum value of the function it defines.

CCSS.MATH.CONTENT.HSA.SSE.B.3.C (../../../../MATH/CONTENT/HSA/SSE/B/3/C/)

Use the properties of exponents to transform expressions for exponential functions. *For example the expression* 1.15^{t} *can be rewritten as* $(1.15^{1/12})^{12t} \approx 1.012^{12t}$ *to reveal the approximate equivalent monthly interest rate if the annual rate is 15%.*

CCSS.MATH.CONTENT.HSA.SSE.B.4 (../../../../MATH/CONTENT/HSA/SSE/B/4/)

Derive the formula for the sum of a finite geometric series (when the common ratio is not 1), and use the formula to solve problems. *For example, calculate mortgage payments.* *